Changes in plasma phospholipid fatty acids and their relationship to disease activity in rheumatoid arthritis patients treated with a vegetarian diet.

❖ ❖ ❖

In a controlled clinical trial we have recently shown that patients with rheumatoid arthritis (RA) improved after fasting for 7-10 d and that the improvement could be sustained through 3.5 months with a vegan diet and 9 months with a lactovegetarian diet. Other studies have indicated that the inflammatory process in RA can be reduced through manipulation of dietary fatty acids. A switch to a vegetarian diet significantly alters the intake of fatty acids. Therefore, we have analysed the changes in fatty acid profiles of the plasma phospholipid fraction and related these changes to disease activity. The concentrations of the fatty acids 20:3n-6 and 20:4n-6 were significantly reduced after 3.5 months with a vegan diet ($P < 0.0001$ and $P < 0.01$ respectively), but the concentration increased to baseline values with a lactovegetarian diet. The concentration of 20:5n-3 was significantly reduced after the vegan diet ($P < 0.0001$) and the lactovegetarian diet periods ($P < 0.01$). There was no significant difference in fatty acid concentrations between diet responders and diet non-responders after the vegan or lactovegetarian diet periods. Our results indicate that the changes in the fatty acid profiles cannot explain the clinical improvement.